CLASS-8

LESSON-13 INTRODUCTION TO GRAPHS

(This PDF Based on NCERT Book)

A LINE GRAPH(एक रेखा ग्राफ)-

A **line graph** (or line chart) is a type of chart used to visualize information that changes over time. It is created by connecting a series of data points, called **markers**, with straight line segments.

It is one of the most effective tools for identifying trends, patterns, and fluctuations in continuous data.

Anatomy of a Line Graph

To read or create a line graph, you need to understand its four primary components:

- **The X-Axis (Horizontal):** Usually represents the **independent variable**, most commonly time (days, months, years).
- **The Y-Axis (Vertical):** Represents the **dependent variable**, which is the quantity being measured (temperature, price, population).
- **Data Points:** Specific values plotted at the intersection of the X and Y coordinates.
- **The Line:** The connection between points that reveals the "story" of the data—whether it is increasing, decreasing, or staying steady.

When to Use a Line Graph

Line graphs are the preferred choice in specific scenarios:

1. **Tracking Changes Over Time:** For example, monitoring how a company's stock price fluctuates throughout a fiscal year.
2. **Comparing Multiple Groups:** You can plot multiple lines on the same graph (using different colors) to compare how two or more subjects perform under the same time constraints.
3. **Identifying Trends:** They make it very easy to see if a value is generally trending upward (growth) or downward (decline).

How to Interpret the Slope

The angle and direction of the line segments provide immediate insight into the data:

Line Direction	**Meaning**
Upward Slope	The value is increasing. A steeper slope indicates a faster rate of change.

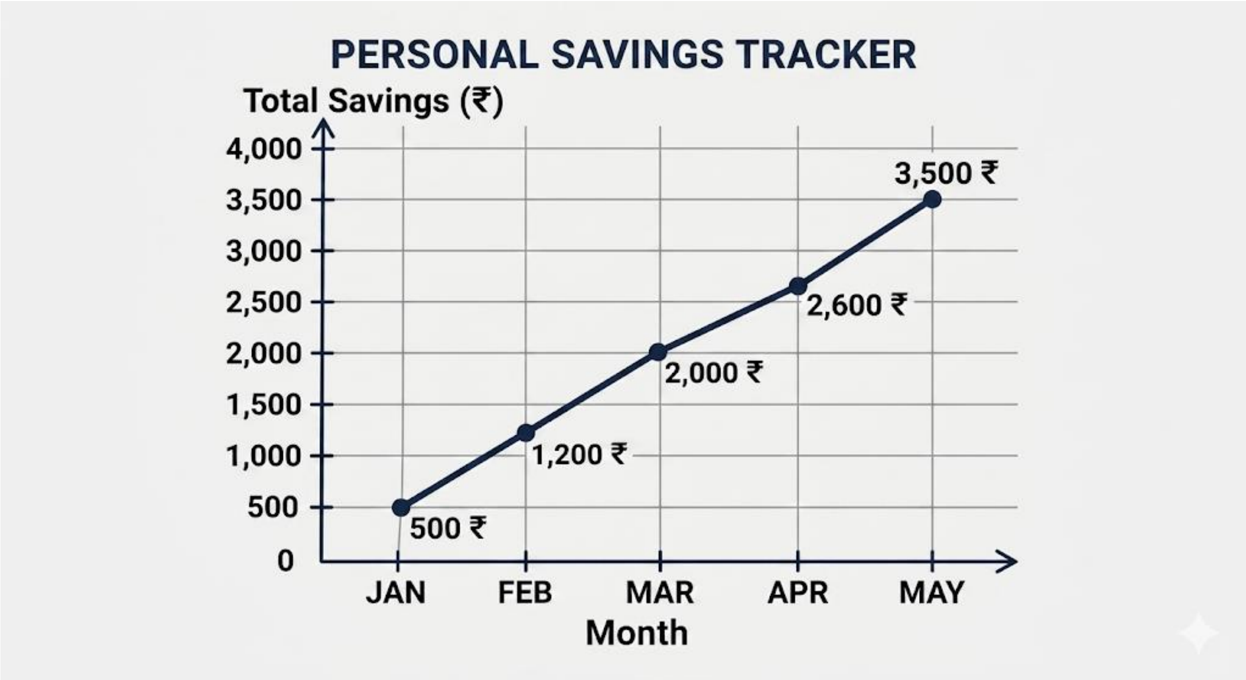
Line Direction	Meaning
Downward Slope	The value is decreasing.
Horizontal Line	There is no change in the value over that period.
Zig-Zag Pattern	Indicates high volatility or frequent fluctuations in the data.

1. Simple Growth (Personal Savings)

This is a great starting point to show a steady upward trend. Imagine you are saving a fixed amount of money every month.

Month	Total Savings (₹)
January	500
February	1,200
March	2,000
April	2,600

Month	Total Savings (₹)
May	3,500



SOME APPLICATIONS(कुछ अनुप्रयोग)-

1. Business and Finance

- **Stock Market:** Tracking the price of a stock (like Google or Apple) minute-by-minute or year-by-year.
- **Sales Tracking:** A shop owner using a line graph to see which months they make the most money.
- **Budgeting:** Comparing your monthly income against your expenses to see if you are saving enough.

2. Science and Health

- **Patient Monitoring:** Doctors use line graphs to track a patient’s heart rate or blood pressure over several hours in a hospital.
- **Climate Change:** Scientists plot global temperatures over the last 100 years to show how the Earth is warming.
- **Chemical Reactions:** Tracking how the temperature of a liquid changes as it is heated in a lab.

3. Education and Sports

- **Grade Progress:** A student tracking their exam scores throughout the year to see if their study habits are working.

- **Fitness Tracking:** Runners use line graphs to see if their speed is increasing over several weeks of training.
- **Match Analysis:** In cricket or football, "Worm" graphs (a type of line graph) show the cumulative runs or goals scored over time.

4. Daily Life and Environment

- **Weather Forecasting:** Checking how the temperature will rise and fall from morning to night.
- **Data Usage:** Your phone settings often show a line graph of how much mobile data you have used throughout the month.
- **Electricity Consumption:** Power companies provide graphs showing when you use the most electricity (e.g., more during the hot afternoon for AC).

Summary Table: What the "Line" Tells You

Application	If the line goes Up	If the line goes Down
Savings	You are getting richer.	You are spending your savings.
Medicine	Fever is increasing.	The patient is recovering.
Business	Profit is growing.	Sales are dropping.
Internet	High data usage.	Low data usage.